

2023 Ph H2 Q4

Section: Our Dynamic Universe

Topic: Relativity

Summary of Question:

Two trains move along parallel tracks: Train A at 3.5 m s^{-1} , Train B at 4.0 m s^{-1} .

(a) Relative speed

$$v_{\text{rel}} = v_{\text{B}} - v_{\text{A}} = 4.0 - 3.5 = 0.5 \text{ m s}^{-1}$$

Answer: 0.5 m s^{-1}

(b) Passenger speed relative to B

$$\text{Passenger relative to platform: } 3.5 - 1.3 = 2.2 \text{ m s}^{-1}$$

$$\text{Relative to B: } 2.2 - 4.0 = -1.8 \text{ m s}^{-1} \rightarrow \text{magnitude } 1.8 \text{ m s}^{-1}$$

Answer: 1.8 m s^{-1}

(c)(i) Speed of emitted light

Speed of light is constant for all observers.

Answer: $3.0 \times 10^8 \text{ m s}^{-1}$

(c)(ii) Length contraction

$$L = L_0 \sqrt{1 - v^2/c^2} = 142 \times \sqrt{1 - 0.9^2}$$

$$L = 142 \times \sqrt{0.19} = 142 \times 0.436 = 61.9 \text{ m}$$

Answer: 61.9 m

Final Answers

(a) 0.5 m s^{-1}

(b) 1.8 m s^{-1}

(c)(i) $3.0 \times 10^8 \text{ m s}^{-1}$

(c)(ii) 61.9 m

Revision Tips

- Use relative velocity addition at low speeds.
- At relativistic speeds, Einstein's velocity addition law is needed.
- Postulate: speed of light is constant in all frames.
- Length contraction only along direction of motion.